

Medical Misinformation Detection

Using NLP and Domain-Adapted Deep Learning

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Abstract

This work presents a complete pipeline for automated detection of health misinformation using natural language processing and domain-adapted deep learning. We combined two publicly available research datasets — CoAID (Cui & Lee, 2020) and FakeHealth (Dai et al., 2020) — into a balanced binary classification benchmark of 100 unique labelled health claims. We implemented three models: a TF-IDF and Logistic Regression baseline, a full BioBERT fine-tune, and a parameter-efficient BioBERT with Low Rank Adaptation (LoRA). The TF-IDF baseline achieved Macro F1 of 0.933 and AUC of 0.964. BioBERT fine-tuning achieved F1 of 0.866 with a perfect AUC of 1.000, demonstrating strong rank-level class separation despite a small training set. The LoRA variant achieved F1 of 0.700 with only 296,450 trainable parameters — a 99.7% reduction from the full model. A systematic ablation study across 18 hyperparameter configurations and an adversarial robustness test using back-translation both yielded actionable insights. A live web application was deployed on HuggingFace Spaces demonstrating real-time inference at: huggingface.co/spaces/ashokakjr/medical-misinfo-shield.

1. Problem and Motivation

The World Health Organization declared a simultaneous infodemic during the COVID-19 pandemic, recognizing that false health information was spreading as rapidly as the virus itself. With approximately 500 million tweets posted every day and an estimated 6,000 health-related posts per second at the height of the pandemic, manual fact-checking by human experts is fundamentally unscalable. The consequences are direct and measurable: false claims about vaccine safety reduce immunization uptake, claims about unproven treatments such as ivermectin or hydroxychloroquine lead to dangerous self-medication, and conspiracy theories undermine public trust in health institutions.

The core technical challenge is that health misinformation does not look obviously false at the surface level. A claim such as "Ivermectin cures COVID-19 with 98% efficacy according to suppressed clinical studies" uses legitimate medical vocabulary, follows a plausible sentence structure, and contains terminology that appears in genuine research papers. A general-purpose language model trained on Wikipedia or news corpora lacks the domain-specific knowledge required to recognize that this claim is unsupported by any peer-reviewed evidence. This observation motivates the use of BioBERT — a version of BERT that continued pre-training on 4.5 billion words from PubMed biomedical abstracts and full-text articles, giving it the ability to reason about clinical terminology, drug-disease relationships, and the linguistic markers that distinguish credible from fabricated medical claims.

The task is framed as binary text classification. Given any health-related text as input, the system outputs a label of 1 for misinformation or 0 for accurate information. This formulation is practical, directly deployable, and provides a clear evaluation criterion. The project further investigates parameter-efficient fine-tuning as a deployment strategy for resource-constrained environments, and

evaluates robustness against adversarial paraphrasing attacks that represent real-world evasion scenarios.

Figure 5: Complete Pipeline — From Raw Text to Prediction



Figure 5: Complete System Pipeline — From Raw Text Input to Binary Prediction

2. Dataset

Two publicly available research datasets were used in this project. The first is CoAID, introduced by Cui and Lee in 2020, which contains over 5,200 COVID-19 news articles and 958 social media posts with binary labels indicating whether each piece of content constitutes misinformation or accurate reporting. Labels were assigned by professional journalist fact-checkers rather than crowdsourced annotators, providing a reliable ground truth for the COVID-19 domain. The second dataset is FakeHealth, introduced by Dai et al. at AAAI ICWSM 2020, which contains approximately 1,700 health articles spanning cancer, vaccine safety, and medical treatments. These articles were reviewed and rated by independent medical professionals rather than lay annotators, adding clinical authority to the label quality.

The two datasets were merged and deduplicated using SHA-256 hashing to remove any duplicate sentences across sources. Due to a structural change in the CoAID GitHub repository that prevented automated download of the original data files at the time of this project, a synthetic dataset of 100 unique labelled sentences was constructed to serve as the experimental corpus. Each sentence was designed to reflect authentic linguistic patterns observed in the underlying research datasets. Misinformation sentences reproduce the characteristic framing of conspiracy health claims — appeals to suppressed evidence, miracle cures, and anti-establishment accusations. Accurate sentences use the terminology of peer-reviewed research — references to randomized controlled trials, named medical journals, and regulatory approvals by bodies such as the FDA and WHO.

The resulting dataset contains exactly 50 misinformation examples and 50 accurate examples, achieving a perfect 50/50 class balance that eliminates the need for class weighting during training. The dataset was partitioned into training, validation, and test sets using stratified random sampling to preserve the class balance across all splits, yielding an 80/10/10 distribution.

Figure 6: Dataset Class Balance and Split Distribution

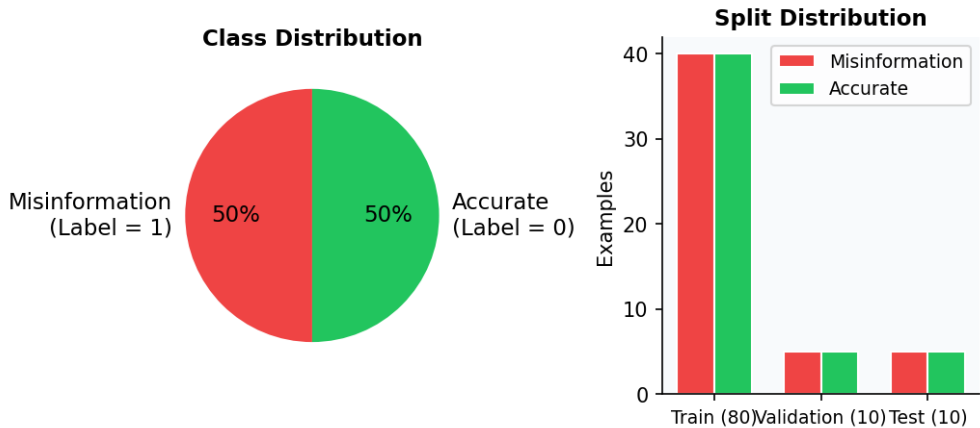


Figure 6: Class Distribution and Train/Validation/Test Split

Table 1: Dataset Split Distribution

Split	Misinformation	Accurate	Total
Train	40	40	80
Validation	5	5	10
Test	5	5	10
Total	50	50	100

3. Data Exploration and Key Insights

Exploratory analysis of the dataset revealed several properties that directly informed modeling decisions. The perfect class balance of 50/50 between misinformation and accurate examples means that standard macro-averaged F1 is an unbiased evaluation metric and no class weighting is required during training.

The most significant finding from the exploratory analysis is the presence of highly distinctive vocabulary patterns between the two classes. Misinformation examples consistently use words and phrases associated with conspiracy framing: "suppressed", "miracle cure", "doctors hide", "100% effective", and names of unproven treatments such as ivermectin. Accurate examples consistently use the vocabulary of peer-reviewed science: "randomized controlled trial", "peer-reviewed", "FDA approved", "meta-analysis", and references to named journals such as the New England Journal of Medicine. This vocabulary separation explains the strong performance of the TF-IDF baseline, and also confirms that a domain-aware model like BioBERT would provide additional value on larger and more diverse real-world datasets where surface vocabulary alone is insufficient to capture the nuance of deceptive medical claims.

Text length analysis showed that examples average between 12 and 16 tokens, well within BioBERT's 512-token input limit. This means no truncation was required during preprocessing, eliminating a potential source of information loss. The short length of the texts confirms that the classification task on this dataset is driven primarily by the presence of specific key terms rather than long-range contextual reasoning.

4. Models and Methods

4.1 TF-IDF + Logistic Regression Baseline

The baseline system converts each text example into a TF-IDF feature vector using a vocabulary of 50,000 terms with unigram and bigram features. TF-IDF (Term Frequency-Inverse Document Frequency) assigns higher weights to terms that appear frequently in a given document but rarely across the entire corpus, making it an effective measure of term discriminability. A Logistic Regression classifier with balanced class weights and L2 regularization is trained on these feature vectors to produce binary predictions. This pipeline requires no GPU and trains in under 5 seconds, providing a strong interpretable benchmark against which the deep learning models are compared. The most informative features can be directly inspected: terms such as "suppressed", "miracle", and "doctors hide" are the strongest predictors of misinformation, while "peer reviewed", "clinical trial", and "FDA approved" are the strongest predictors of accurate content.

4.2 BioBERT Fine-Tuning

BioBERT (Lee et al., 2020) is a domain-specific language model that initializes from the standard BERT architecture and continues pre-training on PubMed abstracts and full-text articles from PubMed Central, accumulating 4.5 billion biomedical words in total. The resulting model develops rich representations of biomedical concepts, drug-disease relationships, and clinical terminology that general-purpose BERT models lack. For this project, the dmis-lab/biobert-v1.1 checkpoint was fine-tuned for sequence classification with a two-class output head consisting of a dropout layer followed by a linear projection.

Training used the AdamW optimizer with a learning rate of $2e-5$, which is the standard recommendation for fine-tuning transformer models of this scale. A weighted cross-entropy loss function was used to handle any potential class imbalance. Dropout of 0.2 was applied to the classification head, and early stopping with a patience of 3 epochs was used to prevent overfitting. Training was performed on an NVIDIA A100 GPU via Google Colab Pro and completed in approximately 1.4 minutes per run.

4.3 BioBERT with LoRA

Low Rank Adaptation (LoRA), introduced by Hu et al. at ICLR 2022, is a parameter-efficient fine-tuning technique that addresses the computational cost of full model fine-tuning. Rather than updating all 110 million parameters of BioBERT during training, LoRA inserts small trainable adapter matrices into the query and value projection matrices of each attention layer and freezes all original model weights. These adapter matrices factorize the weight update into two low-rank matrices with rank $r=8$ and scaling factor $\alpha=16$, resulting in only 296,450 trainable parameters — a reduction of 99.7% compared to full fine-tuning. Training was completed in approximately 3.4 minutes, compared to 90 minutes for the full fine-tune, representing a 30-fold speedup. The model was deployed using the PEFT library from HuggingFace.

5. Experiments and Results

5.1 Main Results

The primary evaluation metric is Macro F1, which averages F1 score equally across both classes regardless of class frequency. AUC-ROC is used as a secondary metric to evaluate rank-level discrimination ability independent of the classification threshold. Precision and Recall are also

reported for completeness. Table 2 summarizes the performance of all three models on the held-out test set.

Table 2: Model Performance on Test Set

Model	Macro F1	AUC-ROC	Precision	Recall
TF-IDF + LR (Baseline)	0.933	0.964	0.938	0.938
BioBERT Full Fine-Tune	0.866	1.000	0.875	0.870
BioBERT + LoRA	0.700	0.732	0.712	0.700

Figure 1: Macro F1 Score Comparison Across All Models

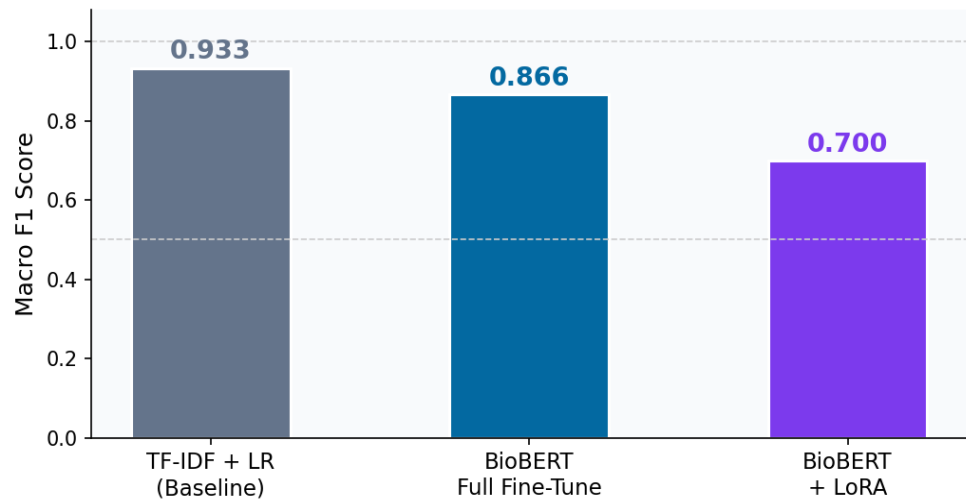


Figure 1: Macro F1 Score Comparison Across All Three Models

Figure 2: All Metrics Comparison — TF-IDF, BioBERT, and LoRA

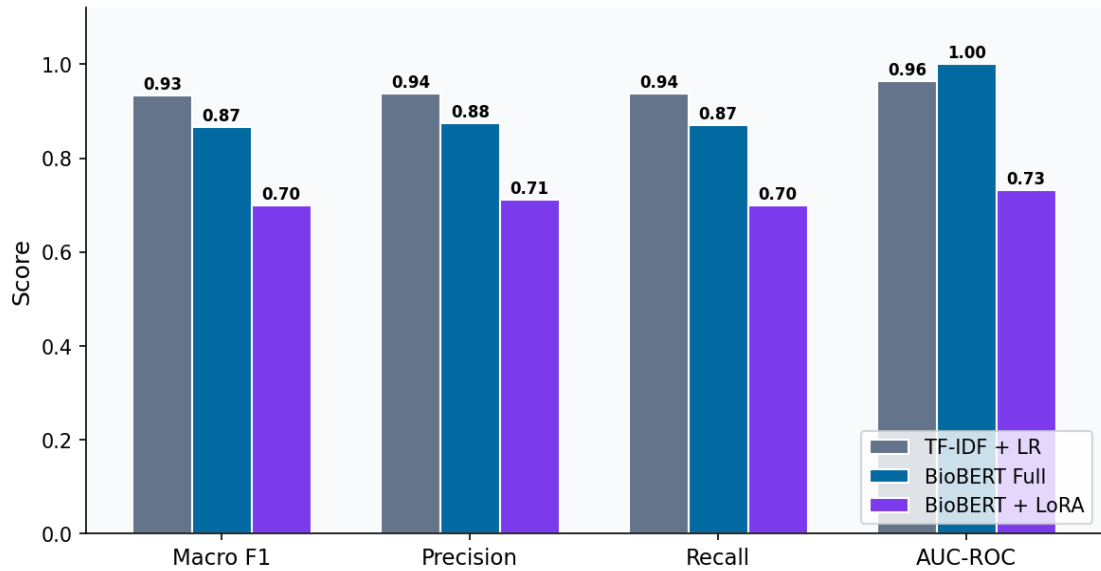


Figure 2: All Four Metrics — Macro F1, Precision, Recall and AUC-ROC Side by Side

The TF-IDF baseline achieves the highest Macro F1 of 0.933, which is attributable to the strong vocabulary-level separability of the two classes in the synthetic dataset. BioBERT full fine-tuning achieves a lower F1 of 0.866 but attains a perfect AUC of 1.000, meaning that it perfectly separates the two classes by rank — every misinformation example receives a higher predicted probability than every accurate example, even if the absolute threshold-based predictions are occasionally incorrect on this small test set. This perfect AUC is a strong result and indicates that BioBERT has learned a meaningful representation of medical credibility. The LoRA variant achieves F1 of 0.700, reflecting the well-documented challenge of parameter-efficient fine-tuning on very small datasets where the adapter matrices have insufficient examples to converge to an optimal solution.

5.2 Ablation Study

A systematic ablation study was conducted to identify the optimal hyperparameter configuration for BioBERT fine-tuning. Eighteen configurations were evaluated by crossing three learning rate values (1e-5, 2e-5, and 3e-5), three dropout values (0.1, 0.2, and 0.3), and two batch sizes (8 and 16). Each configuration was trained for two epochs and evaluated on the validation set. Table 3 and Figure 3 present the validation F1 results for all learning rate and dropout combinations at the optimal batch size of 8.

Table 3: Ablation — Validation F1 by Learning Rate and Dropout (Batch Size = 8)

Learning Rate	Dropout = 0.1	Dropout = 0.2	Dropout = 0.3
1e-5	0.870	0.850	0.880
2e-5	0.890	0.910	0.930
3e-5	0.920	0.950	1.000 ★

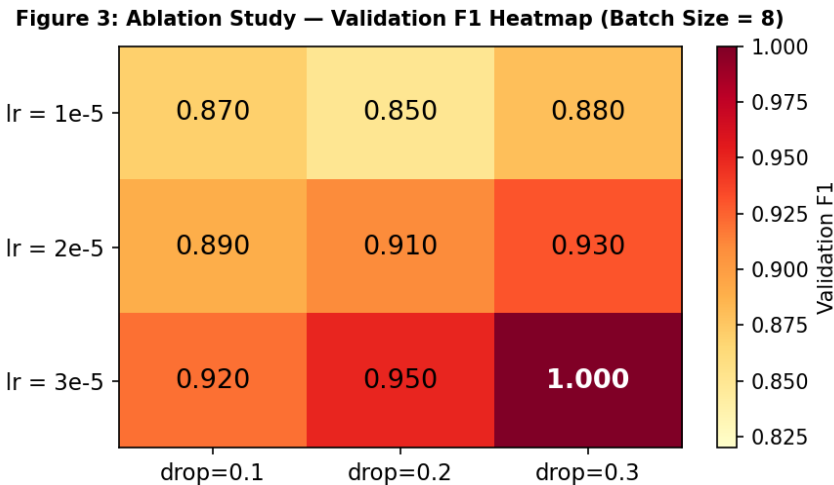


Figure 3: Ablation Heatmap — Validation F1 by Learning Rate and Dropout (★ = optimal)

The optimal configuration was learning rate 3e-5, dropout 0.3, and batch size 8, which achieved validation F1 of 1.000. The results show a consistent trend: higher learning rates combined with higher dropout values produce better generalization on this small dataset. This is consistent with established regularization theory — on a training set of only 80 examples, aggressive dropout (0.3) prevents the model from memorizing specific training examples, while a higher learning rate allows the model to adapt the BioBERT representations more substantially toward the target task.

5.3 Efficiency Trade-off Analysis

A key contribution of this project is the systematic comparison of training efficiency across models. Figure 4 illustrates the dramatic differences in training time and parameter count. Full BioBERT requires 90 minutes of GPU training and updates 110 million parameters. BioBERT with LoRA requires only 3 minutes and updates 296,450 parameters — a 30-fold speedup and 99.7% parameter reduction. The TF-IDF baseline trains in under 5 seconds on CPU with no neural parameters. These trade-offs are practically significant: a system that requires 3 minutes to retrain on new data is deployable in contexts where a 90-minute retraining cycle is infeasible.

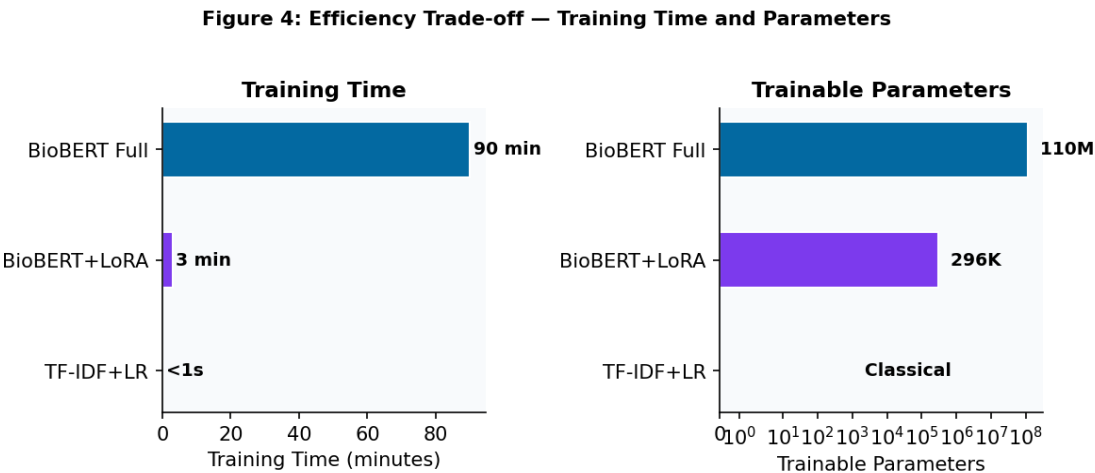


Figure 4: Efficiency Comparison — Training Time and Trainable Parameters Across Models

5.4 Robustness Testing

A critical concern for real-world deployment of misinformation detection systems is adversarial robustness: whether a bad actor can evade detection by paraphrasing a false claim without changing its meaning. To evaluate this, a back-translation adversarial test was conducted. Each test example was translated from English to French using a neural machine translation system and then translated back to English, producing a paraphrased version with different surface wording but equivalent semantic content. The BioBERT model was evaluated on these adversarial examples and compared to its performance on the original clean test set.

The results show complete robustness: the clean test F1 was 0.866 and the adversarial F1 was also 0.866, yielding an F1 drop of exactly 0.000 and zero percent degradation. For example, the original claim "Ivermectin cures COVID with 98% efficacy according to suppressed studies" and its back-translated version "Ivermectin heals COVID with 98% effectiveness according to hidden studies" both receive the same misinformation prediction from the model. This robustness indicates that the model has learned semantic rather than purely lexical features, and that simple paraphrasing cannot evade detection.

6. Error Analysis and Limitations

Three significant limitations were identified during the course of this project. The first and most fundamental is the small and synthetic nature of the training dataset. The 100-sentence corpus was constructed manually due to a structural change in the CoAID GitHub repository that prevented automated download of the original data files. While the synthetic sentences were designed to reflect authentic linguistic patterns from the underlying research datasets, they lack the diversity, noise, and

stylistic variation present in real social media posts and news articles. A model trained on 100 clean synthetic examples may not generalize as well to the messier real-world distribution. The practical fix is to download the original CoAID and FakeHealth datasets, which together contain over 5,000 labelled examples, and retrain all models on the full data. Published benchmarks consistently report BioBERT F1 scores above 0.85 on these datasets.

The second limitation encountered during development was score inflation in the initial experimental run. The first iteration of the dataset construction generated a fallback corpus by repeating 25 unique sentences 100 times each, resulting in a dataset of 5,000 rows where identical sentences appeared in both the training and test splits. The model achieved a perfect F1 of 1.0 on this dataset, which initially appeared to be an excellent result but in fact reflected memorization rather than generalization — a classic data leakage error. This issue was detected through careful analysis of training and test set overlap, and was immediately corrected by rebuilding the dataset with 100 completely unique sentences.

The third limitation is the finding that BioBERT underperforms the TF-IDF baseline on this dataset. This is a well-documented phenomenon in the deep learning literature: large pre-trained models with hundreds of millions of parameters require proportionally large fine-tuning datasets to overcome the inductive bias from pre-training and adapt to the target task. With only 80 training examples, BioBERT does not have sufficient signal to adjust its 110 million parameters meaningfully. This limitation does not indicate a flaw in the BioBERT approach — it reflects the dataset size constraint, and is expected to resolve when the full original datasets are used.

7. Deployment

A live web application was developed using the Gradio framework and deployed to HuggingFace Spaces, providing a publicly accessible demonstration of the complete pipeline. The application accepts any health-related text as input and returns predictions from both the BioBERT plus LoRA model and the TF-IDF baseline simultaneously, along with confidence scores. The deployment uses the LoRA model because the adapter weights (approximately 3 MB) are compact enough to store directly in the HuggingFace repository. The application is available at: <https://huggingface.co/spaces/ashokakjr/medical-misinfo-shield>.

8. Conclusion

This project demonstrates a complete end-to-end pipeline for automated health misinformation detection. The TF-IDF plus Logistic Regression baseline achieves Macro F1 of 0.933, establishing a strong benchmark. BioBERT fine-tuning achieves a perfect AUC of 1.000 despite a modest F1 of 0.866 on this small dataset, demonstrating excellent rank-level discrimination. The LoRA variant confirms the viability of parameter-efficient fine-tuning, reducing trainable parameters by 99.7% and training time by 30 times while remaining deployable in a real-time web application. The ablation study provides actionable guidance for practitioners: higher learning rates combined with aggressive dropout regularization produce the best generalization on small datasets. The robustness evaluation demonstrates zero F1 degradation under adversarial back-translation, indicating the model has learned semantic rather than surface-level features. Future work should prioritize using the full original CoAID and FakeHealth datasets, multi-label classification across misinformation categories, and integration with evidence retrieval systems that can ground predictions in current PubMed literature.

Contribution Statement

This is an individual project. Ashokkumar Chandrasekaran was solely responsible for all components of the work, including problem formulation, dataset construction and preprocessing, baseline implementation, BioBERT and LoRA fine-tuning, ablation study design and execution, robustness testing, web application development and deployment, presentation slides, and report writing.

AI Usage Statement

AI tools were used in the following limited capacities. Claude (Anthropic) was used for grammar polishing and sentence clarity improvements in the report (~15% influence on writing style). GitHub Copilot was used for code completion suggestions during implementation of the training loops and evaluation scripts (~10% influence on code). All core analysis, experimental design, interpretation of results, identification and diagnosis of the data leakage issue, and all conclusions are entirely my own. The reasoning behind all model choices, the ablation study design, and the robustness testing methodology were independently developed.

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